

Sarosij Bose

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EDUCATION

University of Calcutta, Technology Campus

2019-Present

Bachelor of Technology (BTech.) in Computer Science and Engineering

GPA(Incl. 5th semester: 8.58/10)

PUBLICATIONS

Sarosij Bose, Amlan Chakrabarti, **A Fusion Architecture for Human Activity Recognition.**

Accepted at 18th IEEE India Council International Conference (INDICON) 2021. [[Paper](#)] [[Slides](#)] [[Code](#)]

Sarosij Bose, Avirup Dey, **ResCNN: An alternative implementation of Convolutional Neural Networks.**

Accepted at 8th IEEE Uttar Pradesh International Conference (UPCON) 2021. [[Paper](#)] [[Slides](#)] [[Code](#)]

Sarosij Bose, **Lipschitz Bound Analysis of Neural Networks.**

Under Review at IEEE 13th International Conference on Computing Communication and Networking Technologies (ICCCNT) 2022. [[Paper](#)] [[Code](#)]

EXPERIENCE

Saint Mary's University

June 2022 – Present

MITACS Globalink Intern | Advisor: Prof. Jiju Poovancheri

Halifax, NS

- Designing a **sphere mesh based full body tracking** model for real-time applications.
- Developed a novel algorithm for **fast approximation** of template based models.

Indian Institute of Science (IISc) [[Certificate](#)]

November 2021 – March 2022

Winter Research Intern | Advisor: Prof. Anirban Chakraborty

Remote

- Used **Meta Learning** and **Contrastive Learning** for **Person ReId** and extensively compared them.
- Extensively worked on **Domain Adaptation**, **Adversarial Robustness** and **Knowledge Distillation** and combined them with existing approaches.

Indian Institute of Science (IISc) [[Certificate](#)]

June 2021 – August 2021

Summer Research Intern | Advisor: Prof. Kunal N Chaudhury

Remote

- Worked on estimation and regularization of **Lipschitz Constant** for **Convolutional Neural Networks**.
- Showed that existing **tight bound** estimation algorithms lag behind the **actual** Lipschitz constant by **20x-50x**.

University of Calcutta

Oct 2020 – Ongoing

Undergraduate Researcher | Advisor: Prof. Amlan Chakrabarti

Kolkata, India

- Our proposed **I3D-Xception** late fusion model achieved a **Top-1** accuracy of **87.5%** on **UCF-101**.
- Developed an end-to-end **Knowledge Distillation** framework for video classification in **soccer** videos.

KEY PROJECTS

TSCLite | *Computer Vision: Keras, OpenCV, OpenVINO, Onnx, Python.*

April 2021

- Obtained **96%** validation accuracy using a **lightweight** video traffic sign classification model at **30 fps**.

RobustFreqCNN | *Computer Vision: PyTorch, Python, Robustness.*

Oct 2021

- Generated **RCT maps** on the CIFAR-10 dataset by applying **DCT Transform** over adversarial images.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, Javascript, MATLAB (Basic), Cuda (Basic).

Frameworks/APIs: PyTorch, Tensorflow, Keras.

Libraries/Modules: Matplotlib, Seaborn, Sklearn, OpenCV, Openpose, Sonnet, CVXOPT, MediaPipe (Basic).

ACHIEVEMENTS AND EXTRACURRICULARS

- Recipient of the highly selective **MITACS GRI** Fellowship, consisting of a graduate fellowship worth **\$15000**.
- Selected from **500 participants** worldwide for 6th Summer School of AI 2022 IIIT Hyderabad.
- **Winner** of *AI EntrepreNeural 2021*, conducted by **GES, IIT-Kharagpur** for the ***TSCLite*** submission.
- On-Campus **Runners Up** for Hult Prize 2021. Qualified for **Mumbai Regionals**.